BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 097 · Virtual Memory

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What happens to RAM and ROM contents when power is removed?

Q2. Q = (A AND B) OR C. A=1, B=0, C=1. What is Q?
A. 1
B. 0
C. 2
D. 10

Q3. “A Boolean variable represents any unrestricted paragraph of text.” Correct this. Explain why.

Q4. A program joins strings and takes part of an eight-character word. Correct both claims. Explain why. (a) Joining strings adds their numbers. (b) substring(1,3) always means positions 1 to 3 in every language.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Virtual memory — Secondary storage used as backing space for RAM.

Page — Block of memory moved between physical and secondary storage.

Paging — Transferring memory pages between RAM and backing storage.

Thrashing — Severe slowdown caused by excessive page transfers.

New learning

When RAM is insufficient: More working data may be required than physical RAM can hold at once.

Virtual memory: The operating system uses a reserved area of secondary storage as extra backing space for memory pages.

Moving pages: Pages are moved between RAM and secondary storage as needed; this is substantially slower than accessing RAM.

Performance consequence: Frequent movement between RAM and storage can cause a noticeable slowdown (thrashing).

Worked example

A laptop needs 6 GB for active programs but has only 4 GB available RAM. Explain what virtual memory does and why performance may fall. [4 marks]

1. Identify demand and capacity

2. 6 GB is needed but only 4 GB of physical RAM is available.

3. The OS can use some secondary storage as backing for inactive pages.

4. Accessing secondary storage is slower; swapping pages frequently slows the system. [4]

We Do · Practise with guidance

Guided 1

A laptop opens many browser tabs. Explain why virtual memory might be used. [3]

Worked method: The total working memory demand may exceed RAM [1]; the OS temporarily stores less-used pages on secondary storage [1] and brings them back when needed [1].

Guided 2

Explain why adding RAM could reduce use of virtual memory. [3]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Worked method: More active data fits in physical RAM [1], reducing page transfers to secondary storage [1] and therefore decreasing slow accesses [1].

Guided 3

A student says virtual memory makes a laptop faster than RAM. Evaluate the statement. [4]

Worked method: Virtual memory extends apparent working capacity [1] and can allow more programs to remain open [1], but its secondary-storage backing is slower than RAM [1], so heavy paging reduces performance [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Finish the output message when physical RAM is insufficient.

```
neededGB = 8
freeRAMGB = 6
if neededGB > freeRAMGB then
    ...
endif
```

write

Question 1

State what type of storage is used to provide virtual memory. [1] Source: Original OCR J277-style practice question.

Question 2

Explain two reasons why extensive paging can reduce performance. [4] Source: Original OCR J277-style practice question.

Question 3

A program needs 5 GB but only 3 GB of free RAM is available. Explain the purpose of virtual memory in this scenario. [3] Source: Original OCR J277-style practice question.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

An algorithm only gives reliable results when its inputs are handled as specified.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. RAM loses its contents; ROM keeps its contents because it is non-volatile.

Q2. A — 1

Q3. True or false.

Q4. (a) Joining 'AB' and '12' gives 'AB12', with no added space. (b) For 'COMPUTER', substring(1,3) in OCR ERL gives 'OMP': indexes start at 0 and 3 is the length. Follow each language's rules.

I Do / We Do pages · Guided practice

Q1. The total working memory demand may exceed RAM [1]; the OS temporarily stores less-used pages on secondary storage [1] and brings them back when needed [1].

Q2. More active data fits in physical RAM [1], reducing page transfers to secondary storage [1] and therefore decreasing slow accesses [1].

Q3. Virtual memory extends apparent working capacity [1] and can allow more programs to remain open [1], but its secondary-storage backing is slower than RAM [1], so heavy paging reduces performance [1].

You Do pages · Code First

```
neededGB = 8
freeRAMGB = 6
if neededGB > freeRAMGB then
    print("May need virtual memory")
endif
```

Expected result: May need virtual memory

You Do pages · Independent questions

Q1. Secondary storage, e.g. SSD/HDD [1].

Q2. Pages transferred between RAM and storage [1], requiring additional input/output operations [1]; secondary storage is slower than RAM [1], so applications wait for required data [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Q3. Demand exceeds RAM by 2 GB [1]; some pages can be held in secondary storage [1] and restored to RAM when needed [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: